



ST. MARY'S

GROUP OF INSTITUTIONS GUNTUR

DEPARTMENT OF CSE – CYBER SECURITY.

Project Title:
SECURING DATA WITH BLOCKCHAIN AND AI.

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Securing Data with Blockchain and Ai

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Abstract:

- **Problem:** Data is fragmented across the internet, controlled by different parties who distrust each other. Authorizing and validating data usage is difficult, hindering the creation of "real big data" and powerful AI.
- **Solution:** An architecture for secure data storage, computing, and sharing.

➤ SecNet Components:



Blockchain-based data sharing: Ensures data ownership and enables trusted sharing for real big data creation.



AI-based secure computing platform: Creates intelligent security rules for a more secure cyberspace.



Trusted value-exchange mechanism: Provides economic rewards for data sharing, incentivizing participation and promoting AI performance.

Introduction:

- **Data Importance and Security Concerns:** Data is crucial in the cyber world, but not all data can be public. Sensitive data like patient health information needs protection. Additionally, users often lack control over their data stored by service providers.
- **Proposed Solution: Private Data Centers (PDC) with Blockchain and AI:** This approach aims to secure user data. It uses three techniques:

Blockchain: Manages data sharing with access control. Users define who can access their data, and only authorized users can view it.

AI: Functions as a secure computing platform. It checks if a user has permission to access data before allowing access.

Rewards: Users who share data earn rewards, incentivizing data sharing and improving AI performance.

- **Project Focus:** The project will implement these concepts using medical data sharing as an example.

Literature Survey:

Data security in Cyber-Physical Systems (CPS):

- **Data Security Challenge:** Data security is crucial for CPS and AI algorithms. Large amounts of data from various sources are needed, but this data also requires robust protection. Stronger AI can potentially enhance data security by identifying complex threats.
- **Efforts to Decouple Data and Applications:** The passage discusses two approaches to improve data security in CPS:
 - **Amber Architecture:** This separates user data from web applications, granting users control over their information and enabling a search function.
 - **OpenPDS (Open Personal Data Store):** This acts as a secure virtual space for users to manage their data, isolating it from applications and offering a "SafeAnswer" mechanism to reduce data exposure during queries.

- **Blockchain for Secure Data Management:** Blockchain technology emerges as a promising solution for CPS data security. It offers:
 - **Tamper-proof and Traceable Recording:** Data records cannot be altered, ensuring data integrity and traceability.
 - **Incentive Mechanisms:** Blockchain can incentivize data sharing, promoting collaboration and potentially leading to more secure systems.
- **AI for Enhanced Security:** AI can analyze vast amounts of data to identify patterns and make predictions, potentially improving data security in CPS. The passage highlights the need for large datasets to train robust AI models for better data protection.

Existing system:

- **Ineffectiveness of Encryption:** Traditional encryption methods might not be sufficient to fully secure data from attackers.
- **Lack of User Authorization Verification:** Existing systems might not always verify user access permissions properly.
- Hackers use various techniques to potentially crack encryption or exploit vulnerabilities in the system. This can lead to unauthorized individuals gaining access to sensitive data.

Disadvantages:

- **Vulnerability to Attacks:** Encryption, a common protection method, can be broken by determined attackers, leads to data exposed.
- **Uncontrolled Access:** Existing systems might not always check user permissions thoroughly, allowing unauthorized access to sensitive data.

Proposed system:

- The proposed system enables Private Data Centers (PDC) with Blockchain and AI technique to provide security to user's data.
- Blockchain-based data sharing with ownership guarantee, which enables trusted data sharing and generates hash code (unique). data sharing and generates hash code (unique).

Advantages :

- **Enhanced Data Ownership and Control:**
 1. Blockchain technology ensures users retain ownership of their data.
 2. Users define who can access their data, granting granular control over access permissions.
- **Improved Data Security and Trust:**
 1. Blockchain provides a tamper-proof ledger, guaranteeing data integrity and preventing unauthorized modifications.
 2. Trust in the system is established through secure data sharing mechanisms.

SOFTWARE REQUIREMENTS:

- IDE: PyCharm
- Front End: Html, CSS, JS
- Back End: MySQL Server
- Operating System: Window 7/Window 10
- Programming Language: Python

HARDWARE REQUIREMENTS:

➤ Developing Time Requirements

- Intel ® Core i3 CPU 3.0 GHz or above
- 4GMRAM
- 500GB Hard Disk Drive (HDD) + 24GB Solid State Drive(SSD)
- Monitor

➤ Runtime Requirements

Processor: Intel Atom® Processor Z2520 1.2 GHz, or faster processor Storage: Between 850 MB and 1.2 GB, depending on the language version □ RAM: Minimum of 512 MB, 2 GB is recommended □ Hard Disk: 2GB of available hard-disk space for installation; extra free space is required during installation.

Justification of Platform:

- ❑ Python
- ❑ Django
- ❑ mysql

- **Python:** It's a general-purpose, versatile language with a clear syntax and easy-to-read code. It also offers automatic memory management and supports various programming styles.
- **Django:** This high-level Python web framework is ideal for rapid development and clean design. It provides a foundation for building complex web applications without needing to write everything from scratch.
- **MySQL:** This open-source relational database management system (RDBMS) is popular for its ease of use, speed, and compatibility with various programming languages, including Python. It can handle large datasets efficiently.

Proof of Work (PoW) algorithm:

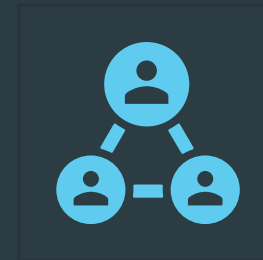
POW algorithm is a cryptographic consensus mechanism used in some blockchains to validate transactions and secure the network.



Mining Process: PoW involves complex puzzles solved by miners using computing power. This creates a secure and verifiable record of transactions on the Blockchain.

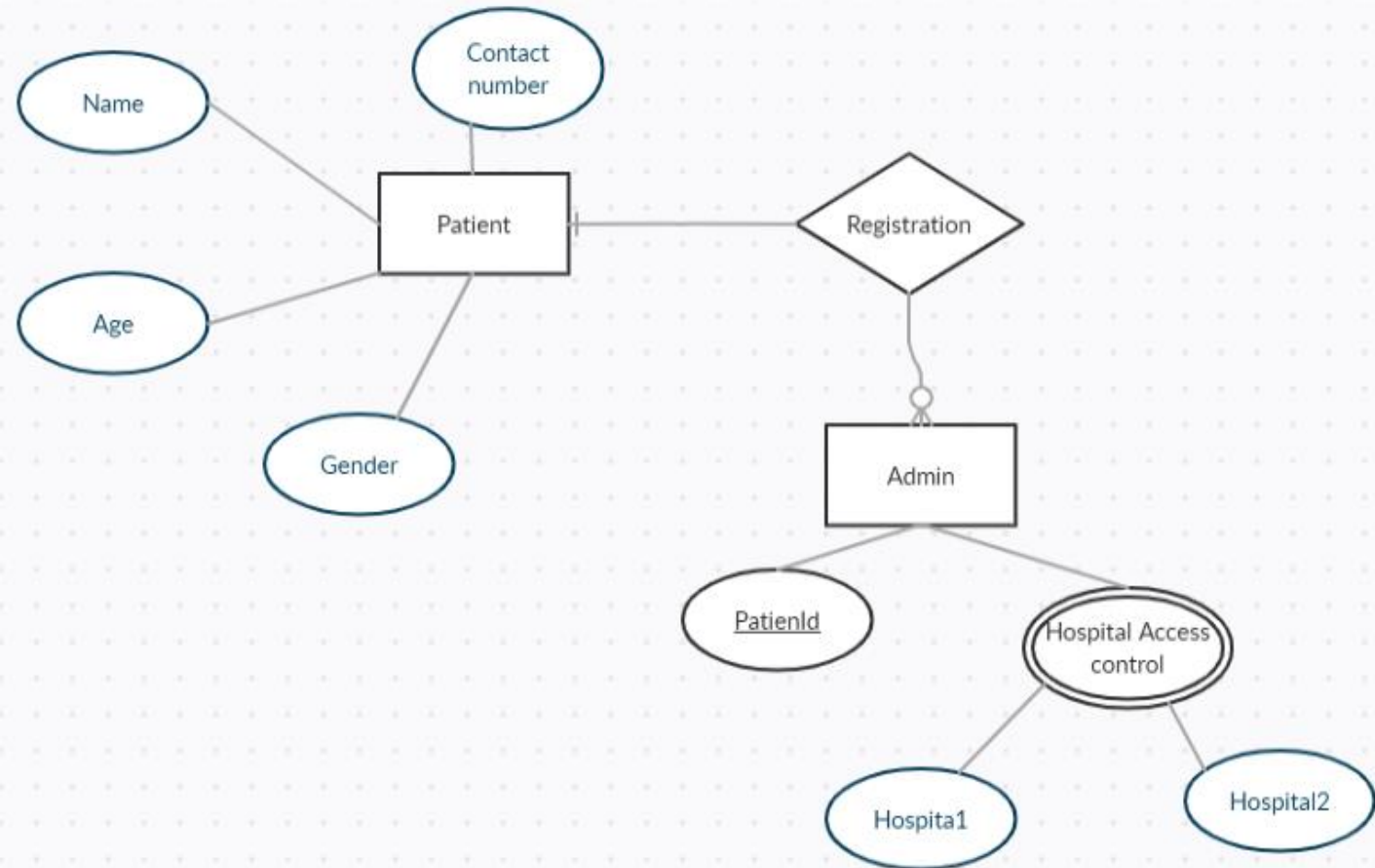
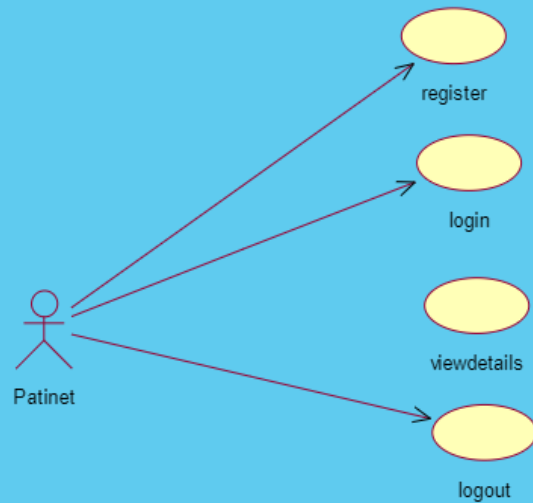
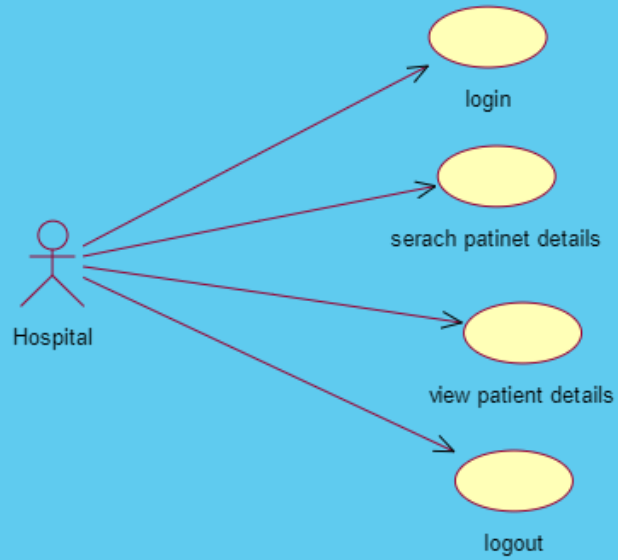


Tamper-proof Ledger: The difficulty of solving these puzzles makes it nearly impossible to alter past transactions on the Blockchain, protecting data integrity.



Network Decentralization: Anyone can participate in mining, which distributes the validation process and prevents a single entity from controlling the network. This strengthens overall security.

E-R diagram representation:



Experimental Results:

➤ Deployment and Functionality:

- You can deploy the code on Django and run the server. The provided figures showcase the functionalities:
- Patients can register, create profiles, and share data with specific hospitals.
- Hospitals can log in, search for patients based on disease, and access data based on granted permissions.

➤ Security and Transparency:

- Blockchain ensures data security and access control.
- Only authorized hospitals can access patient data.
- Patients can see a hash code (unique identifier) for their data and their earned rewards for data access.

Deploy code on DJANGO and start server and run in browser to get below screen

Inbox (171) - kaleem.mmd@gmail.com x Develop a blockchain application x Securing Data With Blockchain and AI x

localhost:8000/index.html

Securing Data With Blockchain and AI

Home Hospital Patients Login New Patient Register Here

FIGURE 2. Medical data sharing using SecNet.

Abstract-Data is the input for various artificial intelligence (AI) algorithms to mine valuable features, yet data in Internet is scattered everywhere and controlled by different stakeholders who cannot believe in each other, and usage of the data in complex cyberspace is difficult to authorize or to validate. As a result, it is very difficult to enable data sharing in cyberspace for the real big data, as well as a real powerful AI.

In this paper, we propose the SecNet, an architecture that can enable secure data storing, computing, and sharing in the large-scale Internet environment, aiming at a more secure cyberspace with real big data and thus enhanced AI with plenty of data source, by integrating three key components:

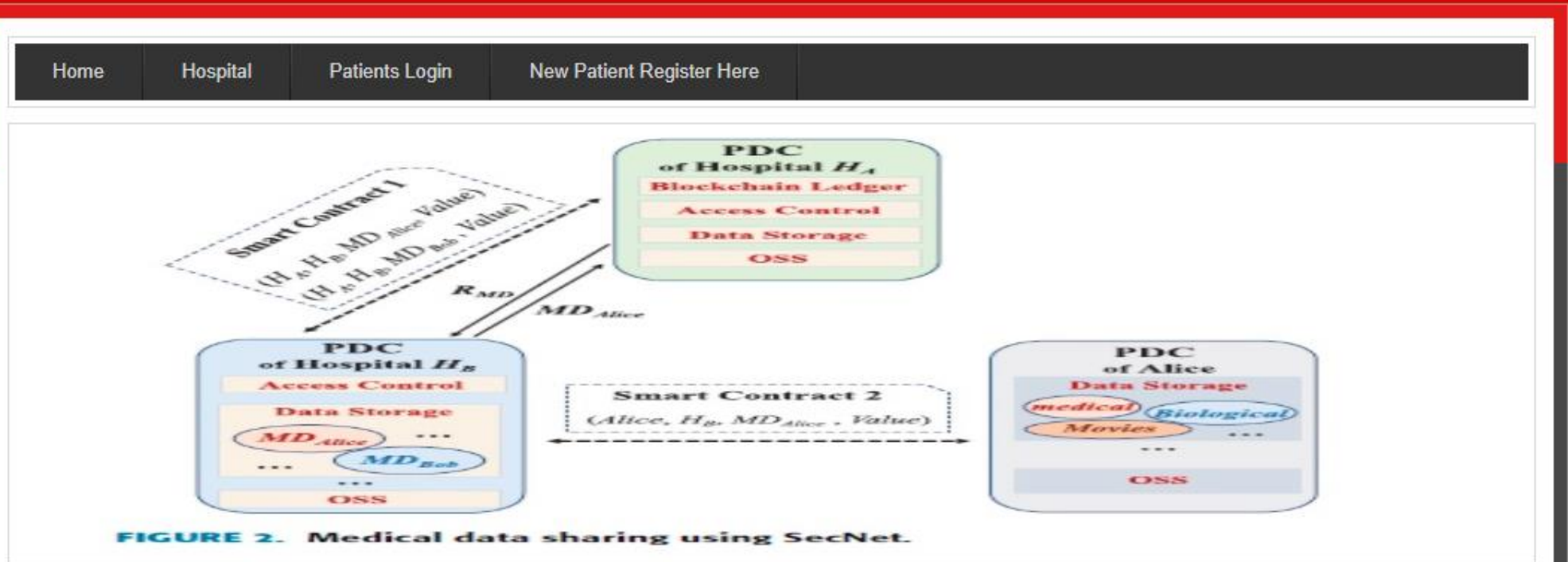
Windows Taskbar: Type here to search, 17:56, 17-12-2019



Patients Profile Creation Screen

Patient Name	himesh
Age	30
Problem Desc	chest pain
Access Control	Hospital 1 Hospital 2
Gender	Male
Contact No	9652861905
address	hyd
Create	

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Patients Profile Creation Screen

Profile Creation Process Completed. Your Patient ID : 1

Patient Name

Age

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Hospital Login Screen

Username Hospital1

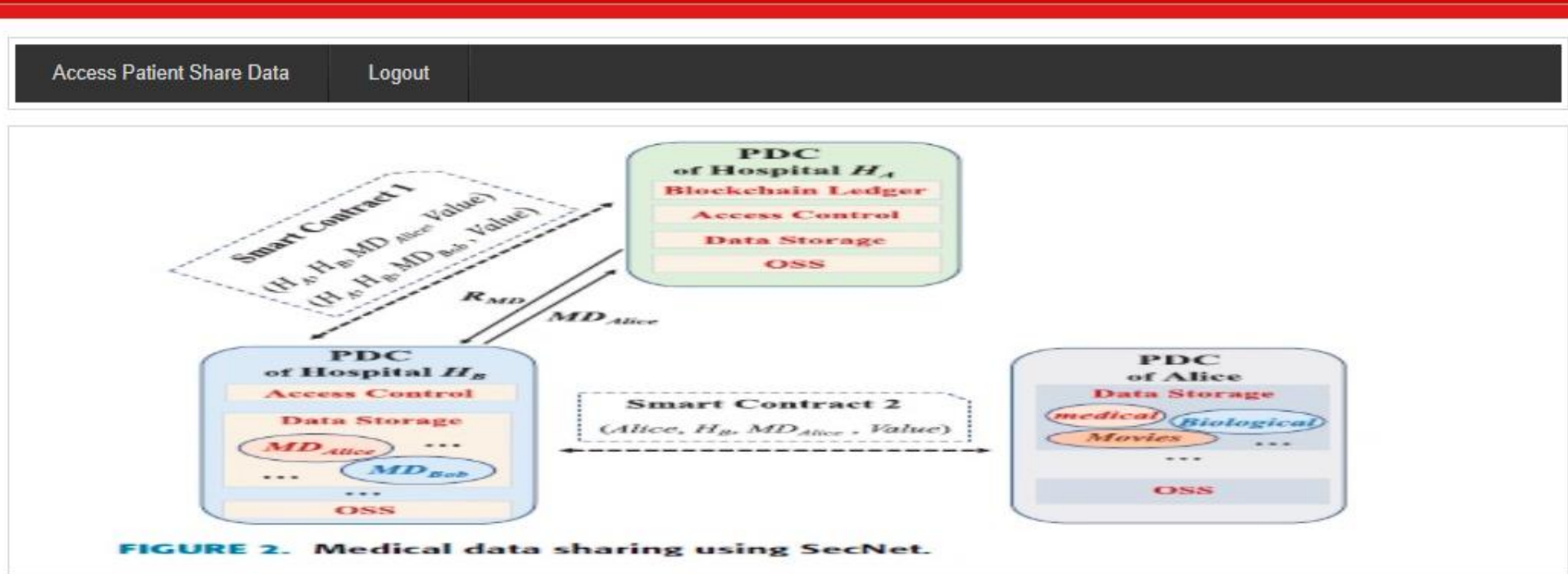
Password

Login

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Patient Share Data Access Screen

AI Search String

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FIGURE 2. Medical data sharing using SecNet.

Patient ID	Patient Name	Age	Problem Description	Profile Date	Access Control	Gender	Contact No
1	himesh	30	chest pain	2019-12-17	Hospital1	Male	9652861905

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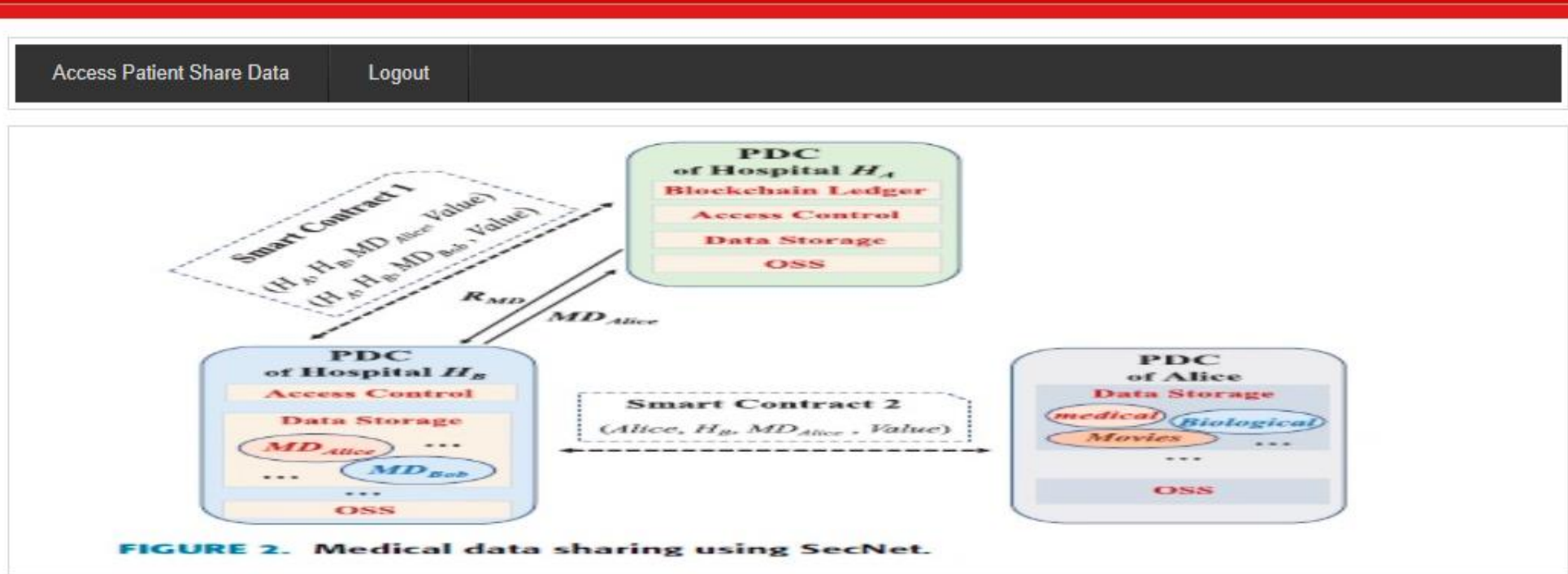
Hospital Login Screen

Username Hospital2

Password

Login

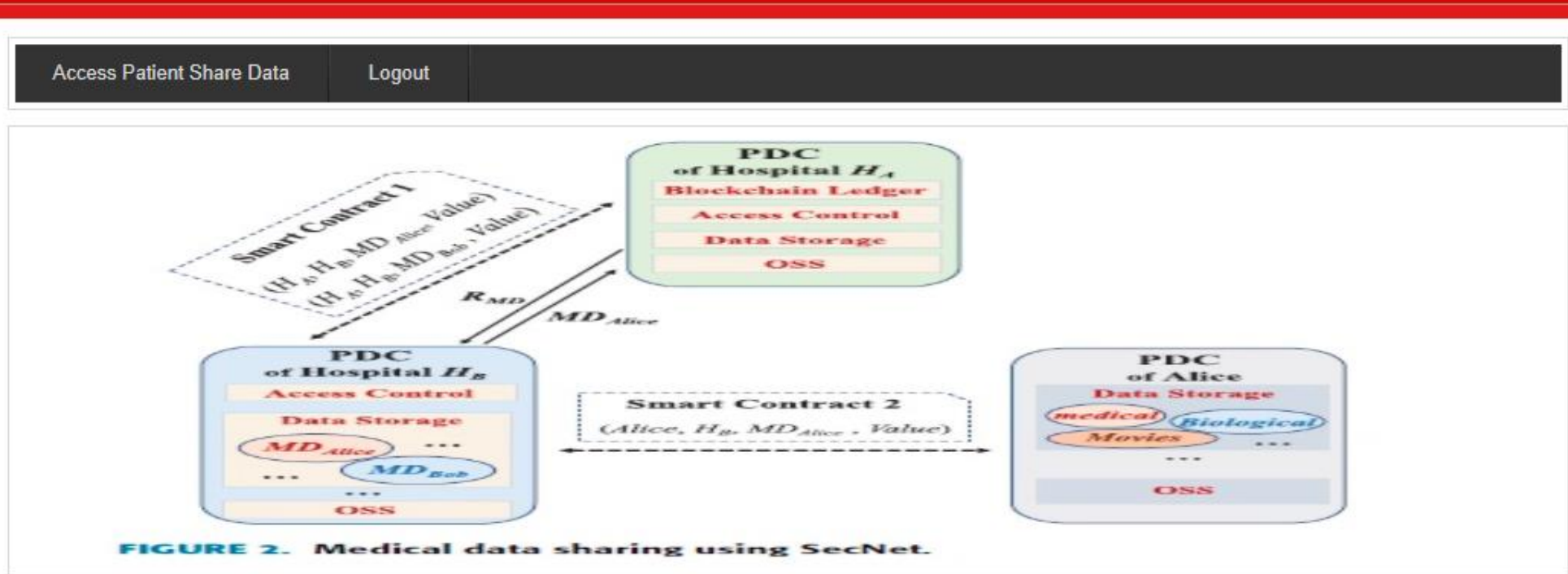
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Patient Share Data Access Screen

AI Search String

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Patient ID	Patient Name	Age	Problem Description	Profile Date	Access Control	Gender	Contact No
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Patient Login Screen

Patient ID

Login

Logout



Patient ID	Patient Name	Age	Problem Description	Profile Date	Access Control	Gender	Contact No	Address	Block Chain Hashcode		Revenue
1	himesh	30	chest pain	2019-12-17	Hospital1	Male	9652861905	hyd	00f3b02362c390c39df03d59e94ee8574a953e1fcd18493e2c93db91c70cf43		0.5

Conclusion:

- **Combating Data Abuse and Empowering AI:** SecNet aims to address data abuse and empower AI in untrusted environments. It focuses on secure data storage, sharing, and computation instead of just communication.
- **Core Technologies:** SecNet leverages three key technologies:



Blockchain:
Guarantees data
ownership and
access control.



AI-based secure
computing
platform:
Analyzes access
requests and
ensures authorized
data usage.



Blockchain-based
incentive
mechanism:
Rewards users for
sharing data,
promoting data
contribution and
fostering a robust
AI.

- **Applications and Transparency:** SecNet's use case is explored in healthcare, but its secure storage functionality can be applied in various scenarios. Patients can view a hash code (unique identifier) for their data and their data access rewards.

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- <https://www.javatpoint.com/blockchain-hash-function>

"Thank you
for your
patience."